New Kernel Functions can be derived From other Kernels

Sum of Kernels is a Kernel

$$K(u, v) = K1(u, v) + K2(u, v)$$

Scalar Product of Kernel is a kernel

$$K(u, v) = \alpha\, K1(u, v)$$

Product of Kernels is a Kernel

$$K(u, v) = K1(u, v) * K2(u, v)$$

Combination of the above rules is a Kernel

$$K(u, v) = K1(u, v) * K2(u, v) + \alpha\, K3(u, v)$$